

Exploratory Data Analysis Report

1. Project Overview

Objective

The objective of this project was to perform Exploratory Data Analysis (EDA) on a cleaned e-commerce dataset to identify patterns, trends, outliers, and actionable business insights.

Dataset Overview

- Total Records: 1,200
- Total Columns: 14
- Dataset Type: E-commerce Transactions

Tools Used

- Python
- Pandas
- Matplotlib
- Seaborn
- Google Colab

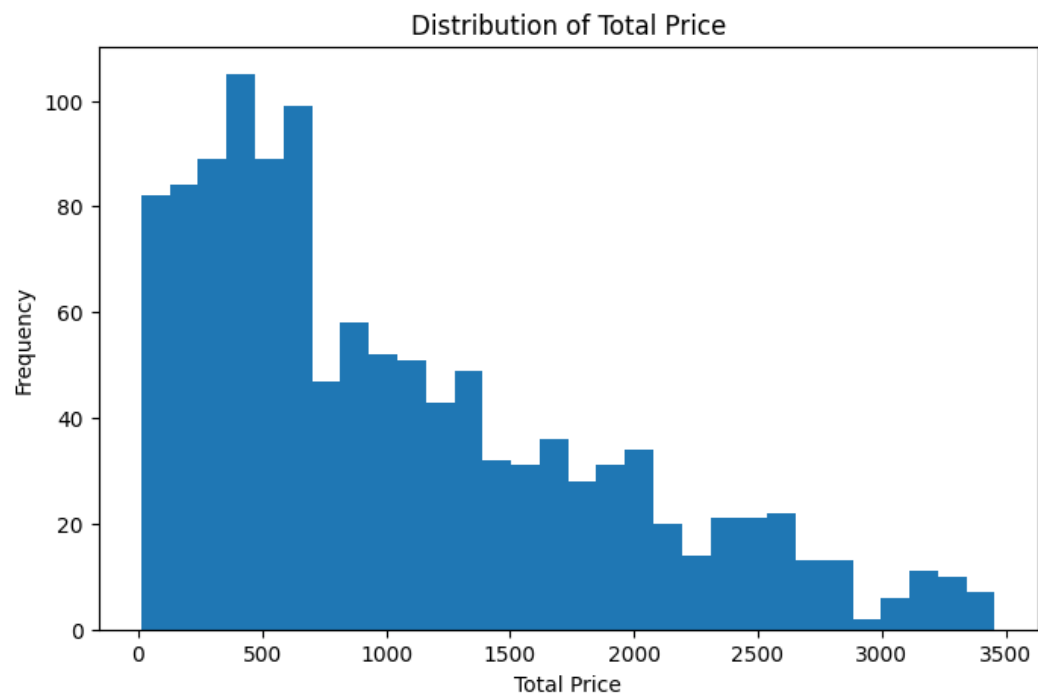
2. Methodology

The following analyses were performed:

- Descriptive Statistics
- Missing Value Verification
- Outlier Detection
- Revenue Distribution Analysis
- Product Performance Analysis
- Payment Method Analysis
- Referral Source Analysis
- Order Status Analysis
- Monthly Revenue Trend Analysis
- Correlation Analysis

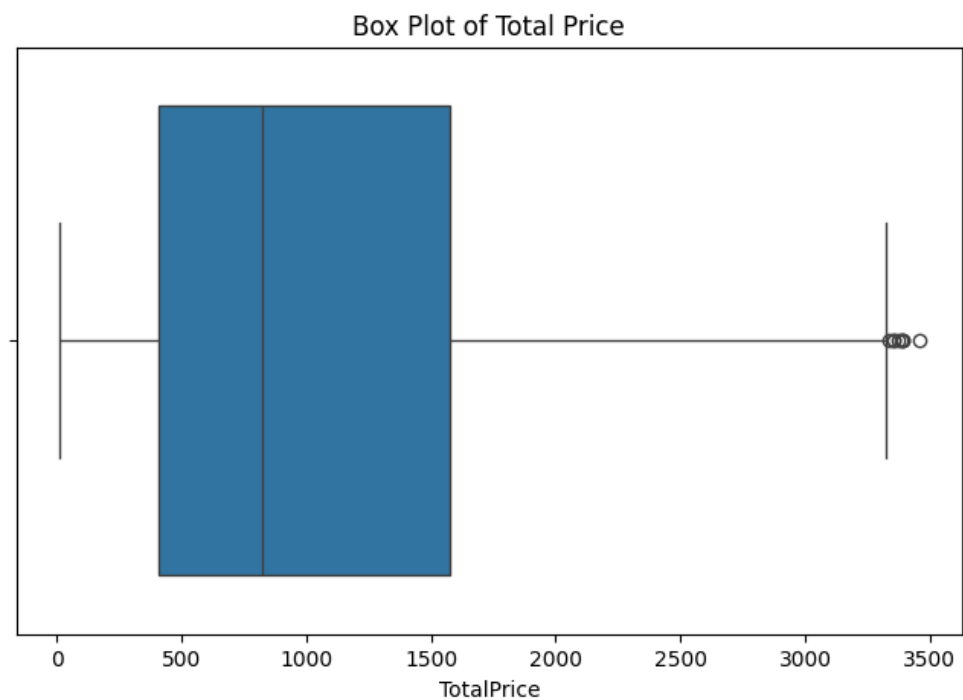
3. Key Visualizations & Findings

Revenue Distribution



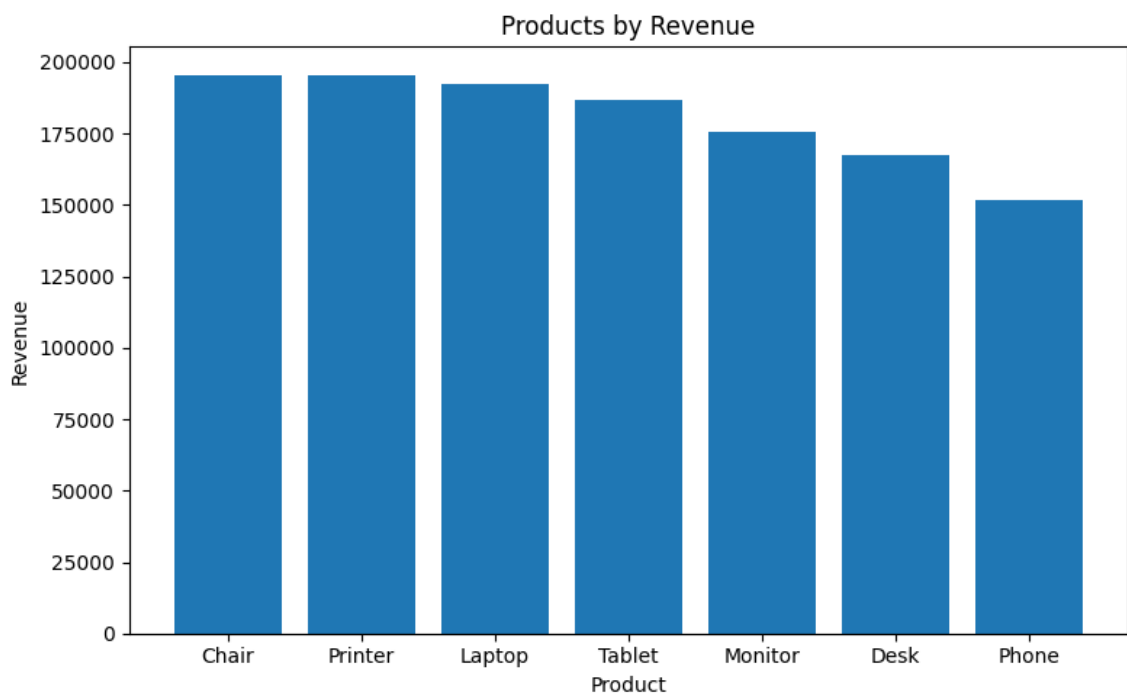
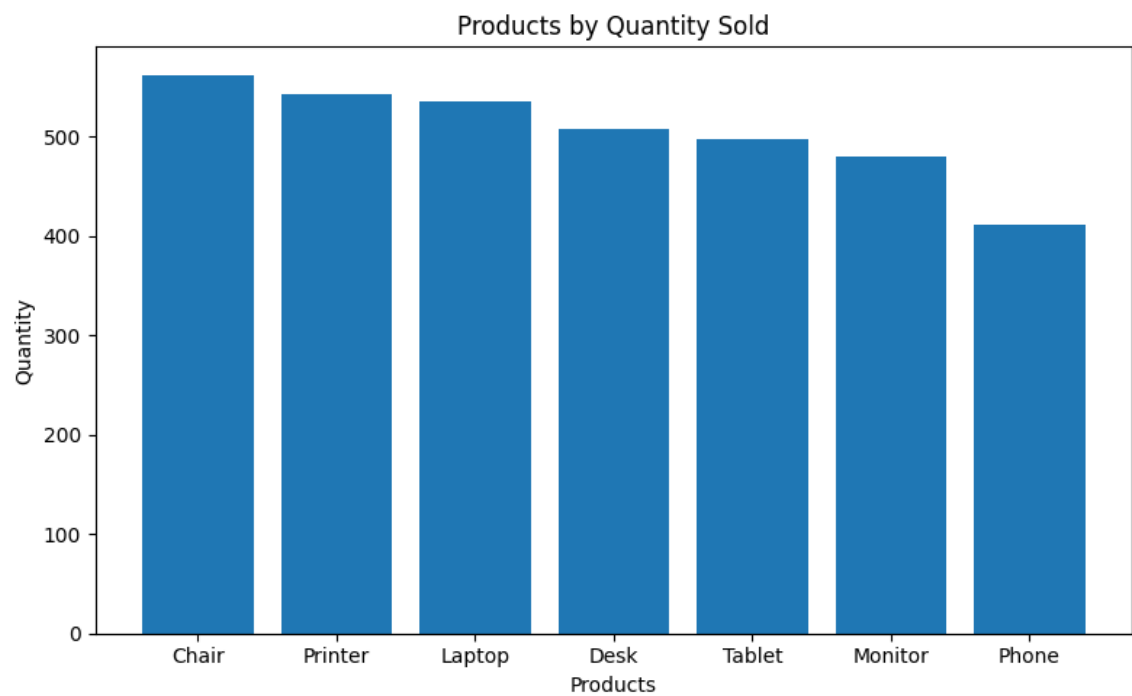
Finding: Most orders were concentrated in lower price ranges, producing a right-skewed distribution.

Outlier Analysis



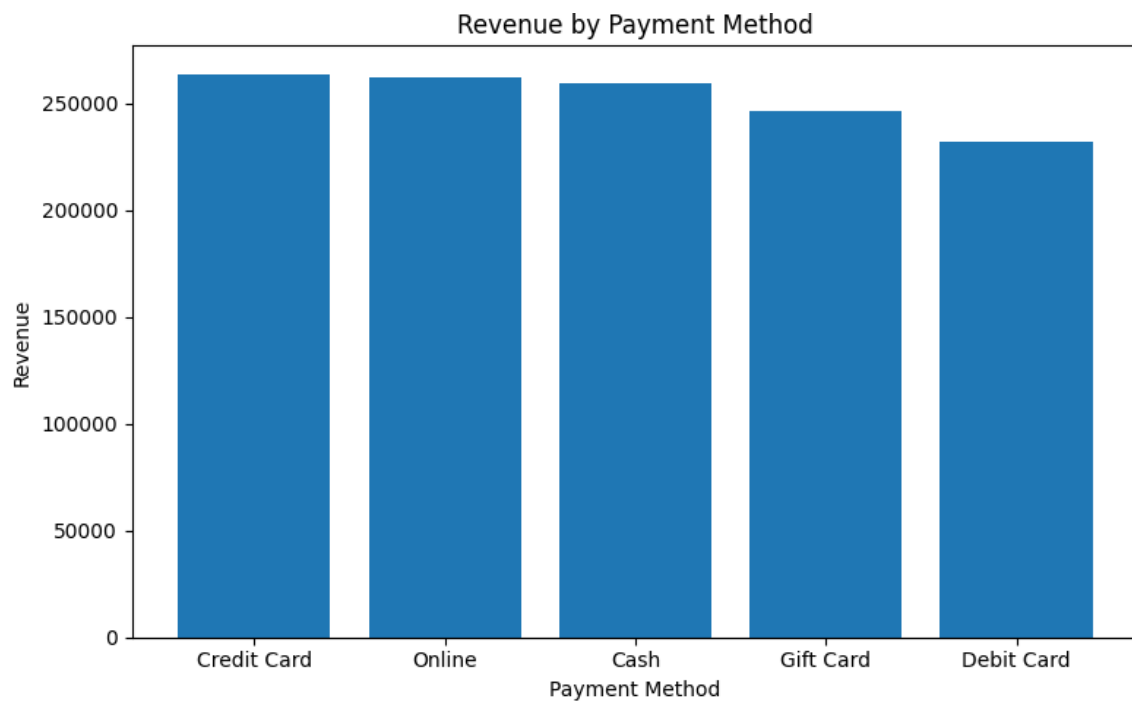
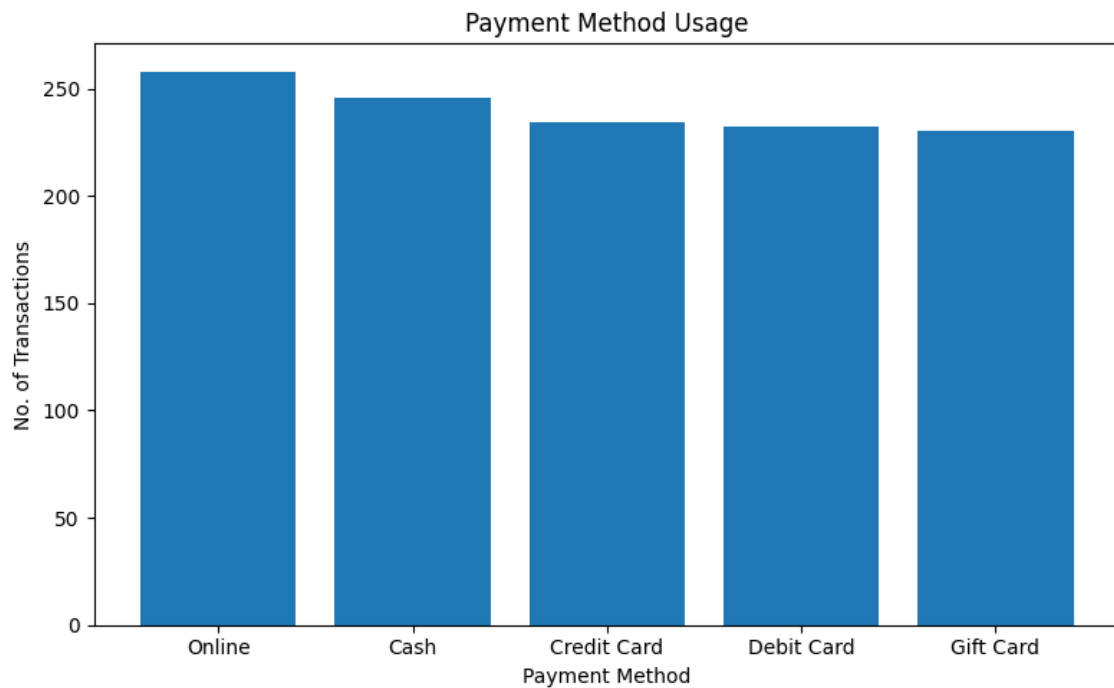
Finding: Several high-value transactions were identified as outliers. Investigation confirmed that these were legitimate business transactions and not data errors.

Product Performance



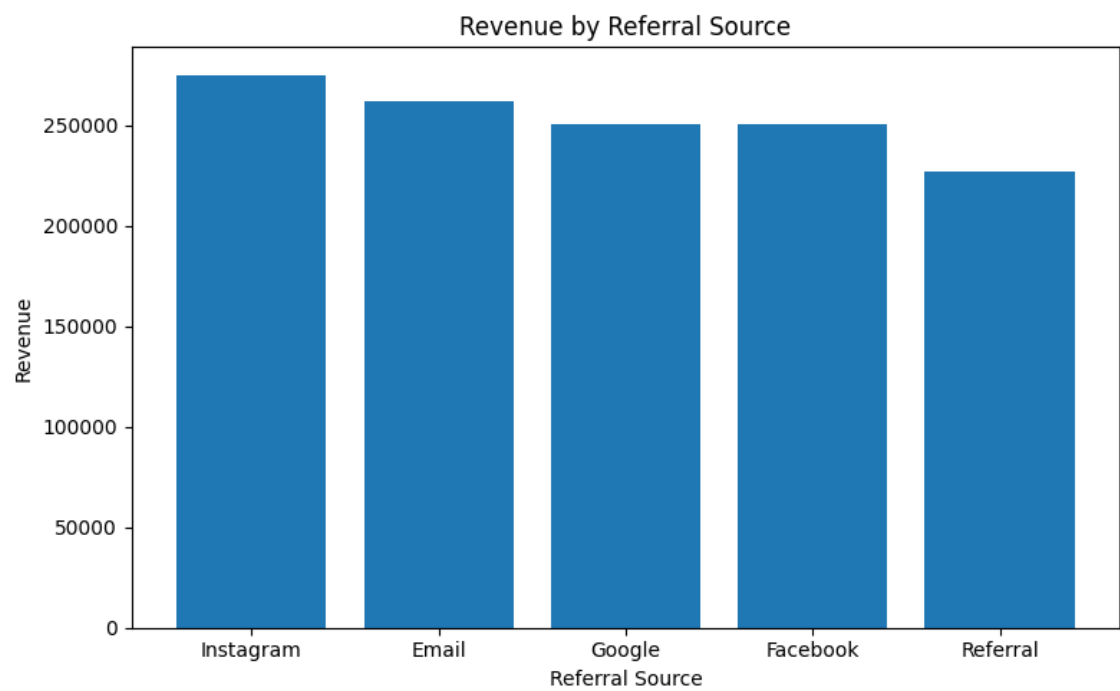
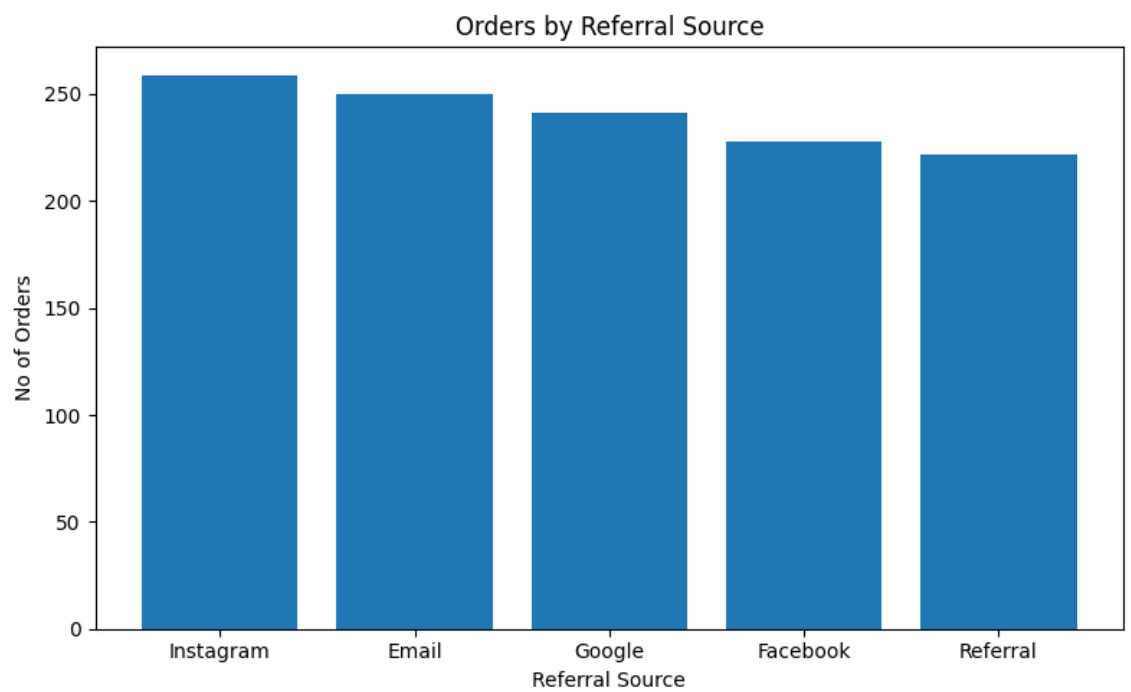
Finding: Chairs generated the highest sales volume and revenue, making them the strongest-performing product category.

Payment Method Analysis



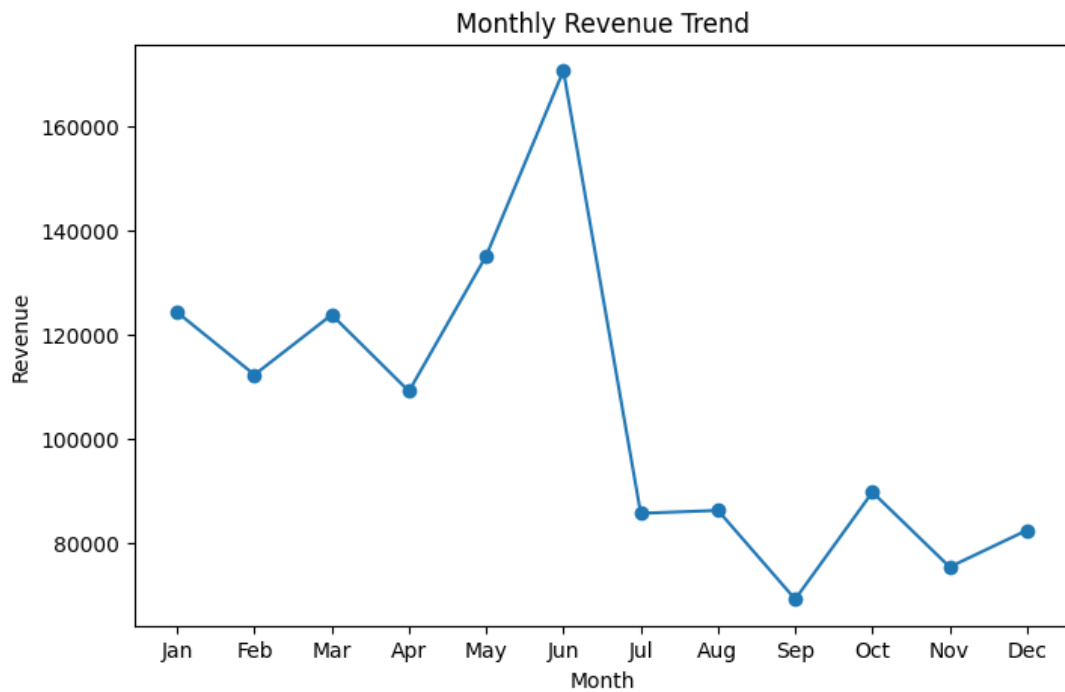
Finding: Online payments were the most common payment method, while Credit Card transactions generated the highest revenue.

Referral Source Analysis



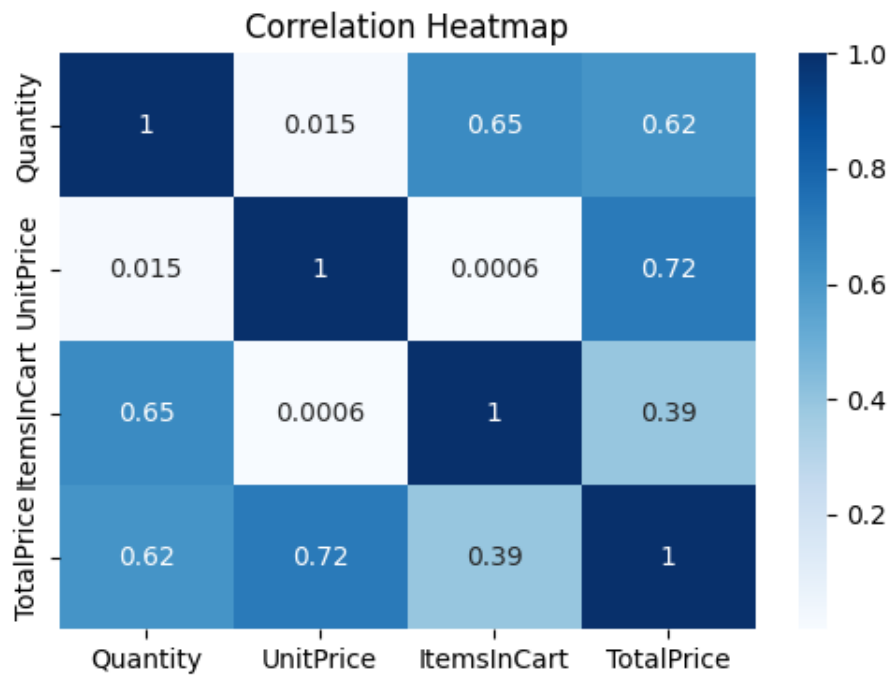
Finding: Instagram generated both the highest number of orders and the highest revenue.

Monthly Revenue Trend



Finding: Revenue peaked in June and declined during the second half of the year.

Correlation Analysis



Finding: UnitPrice and Quantity were the strongest drivers of TotalPrice.

4. Key Insights

- Most sales come from lower-priced orders.
- High-value orders are legitimate and contribute significantly to revenue.
- Chairs are the top-performing product.
- Credit Card users generate the highest revenue.
- Instagram is the most effective acquisition channel.
- Order cancellations represent a significant revenue risk.
- Revenue peaks in June and declines afterward.
- Revenue is primarily driven by product price and quantity purchased.

5. Recommendations

- Grow Instagram marketing investment.
- Investigate and reduce order cancellations.
- Prioritize top-performing products and encourage bundling.
- Manage seasonal revenue fluctuations.
- Encourage credit card usage through incentives.

6. Conclusion

The analysis revealed important patterns in customer purchasing behavior, product performance, marketing effectiveness, and revenue trends. The findings provide actionable insights that can support decision-making in marketing, sales, and customer retention strategies.